



Wav.io: The Brain Cancer Microgravity Modeller

A novel acoustic platform redefining how we model and defeat glioblastoma.

A Complex and Aggressive Enemy

"Martin, 52, died within 4 months of his first headache. Cause: Glioblastoma."

<10%

5-Year Survival

Most patients die within 12 months of diagnosis

~\$1B

Per Drug Approved

Cost of development – wasted on models that don't translate

2D Models

Cells grown on flat plastic – silences the genes that make GBM lethal

3D Models

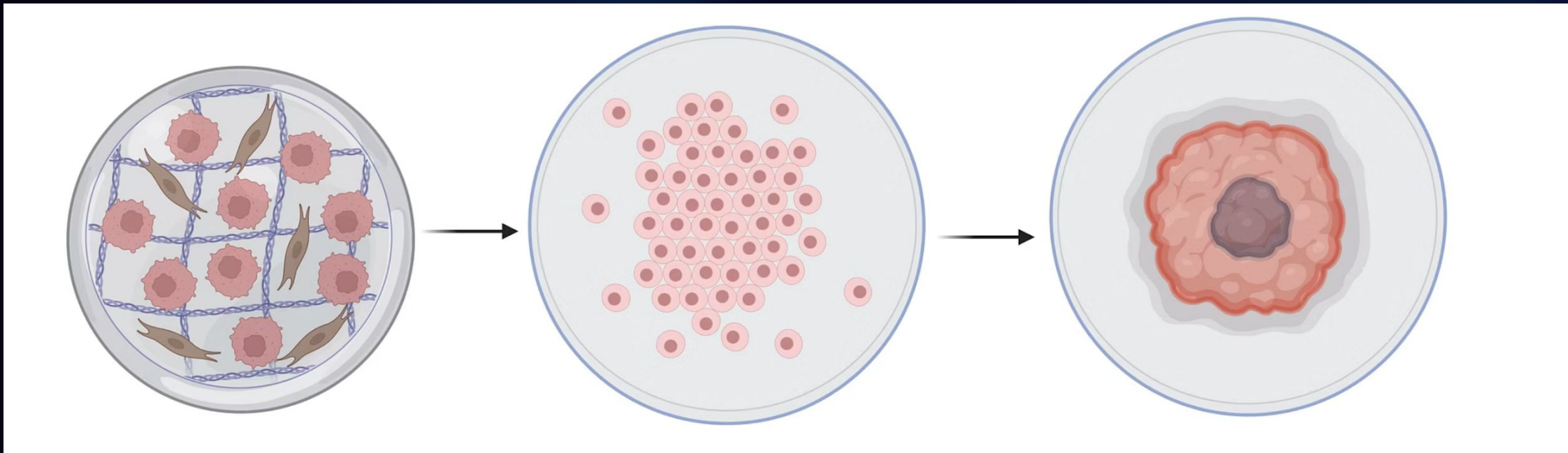
Gravity deforms spheroid shape – inconsistent, unrepresentative

The Result

Drugs pass lab tests and fail in humans. Patients like Martin run out of time.

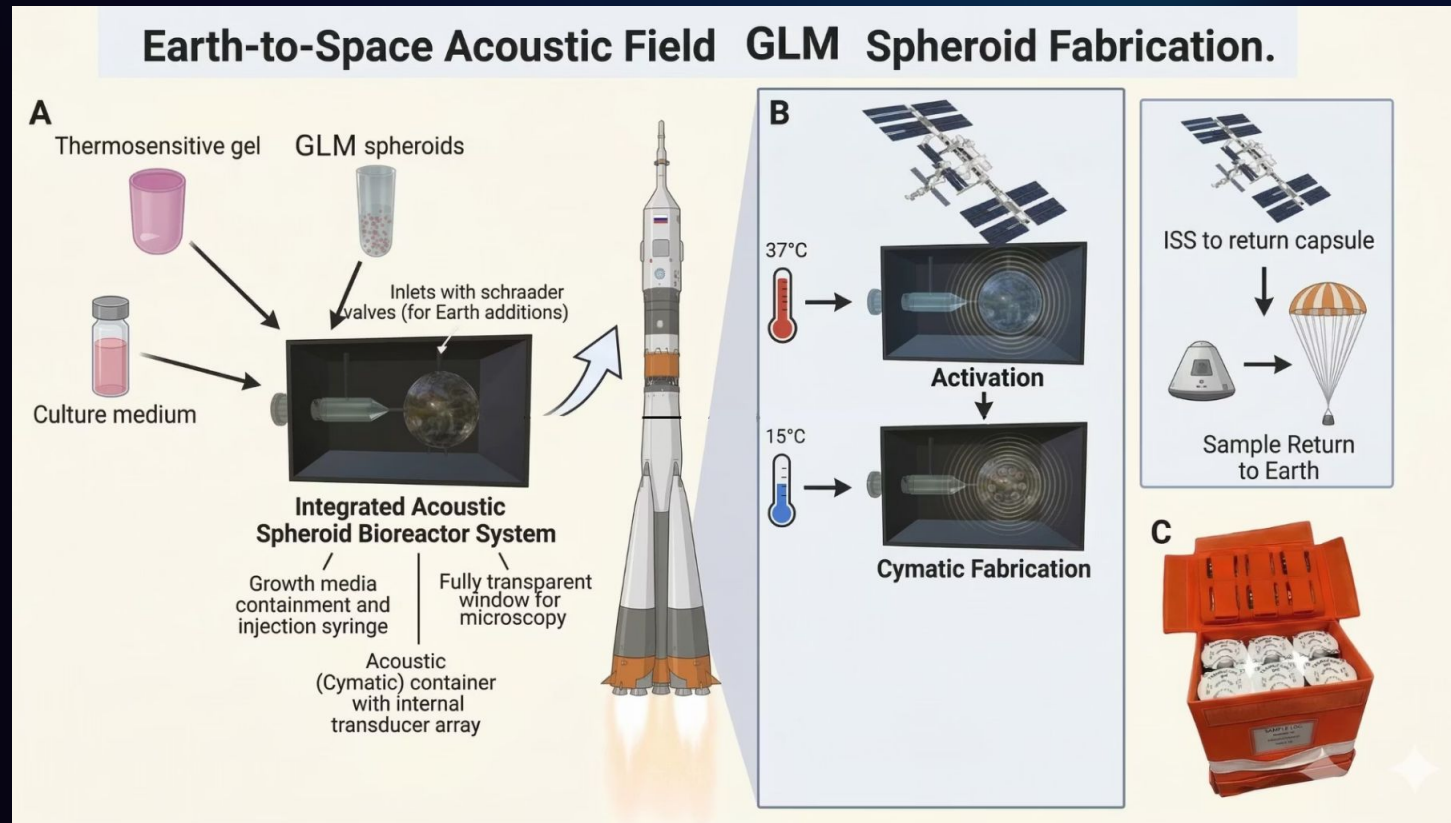


The Insight



Removing gravity eliminates spheroid deformation – but freely floating cells need a new control mechanism. Cymatics, delivered via a precision acoustic transducer array, provides exactly that: sound as a scalpel for spatial organisation.

Our Solution: A Novel Disease Modelling System



01

Launch

0

2 Liquify

0

3 Organise

0

4 Test

A novel rig designed specifically for GBM modelling in microgravity, grounded in state-of-the-art space biology research conducted by V. A. Parfenov et al., "Magnetic levitational bioassembly of 3D tissue construct in space," Science Advances, vol. 6, no. 29, p. eaba4174, 2020, doi:doi:10.1126/sciadv.aba4174.

Live Simulation

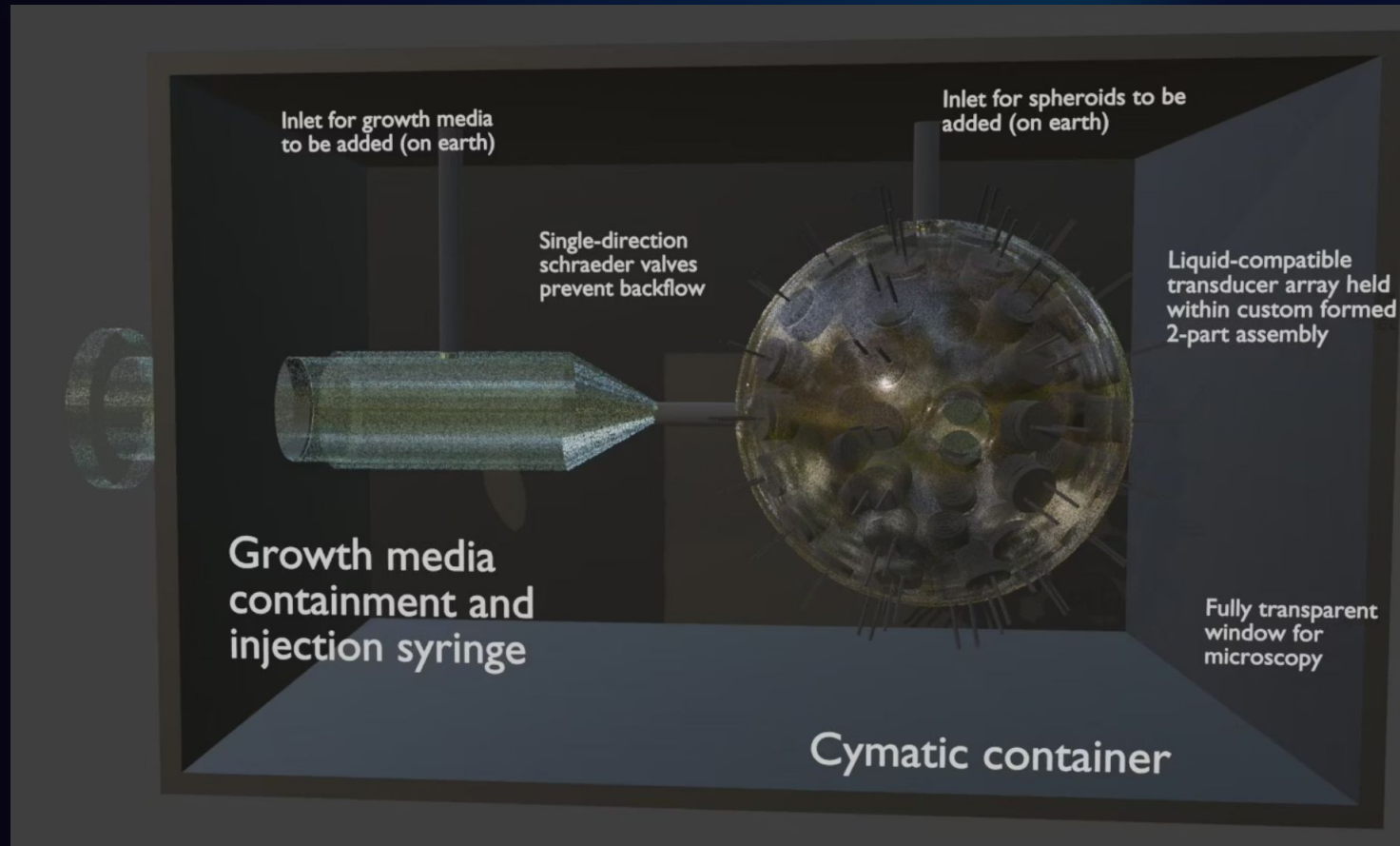
Acoustic Field

Visualised in real-time fluid medium

Spheroid Response

Aggregation dynamics under
phase-controlled pressure

The Rig Design



64 Transducers

Spherical polymeric enclosure – 8 per octant for uniform acoustic field coverage



5 Sealed Orifices

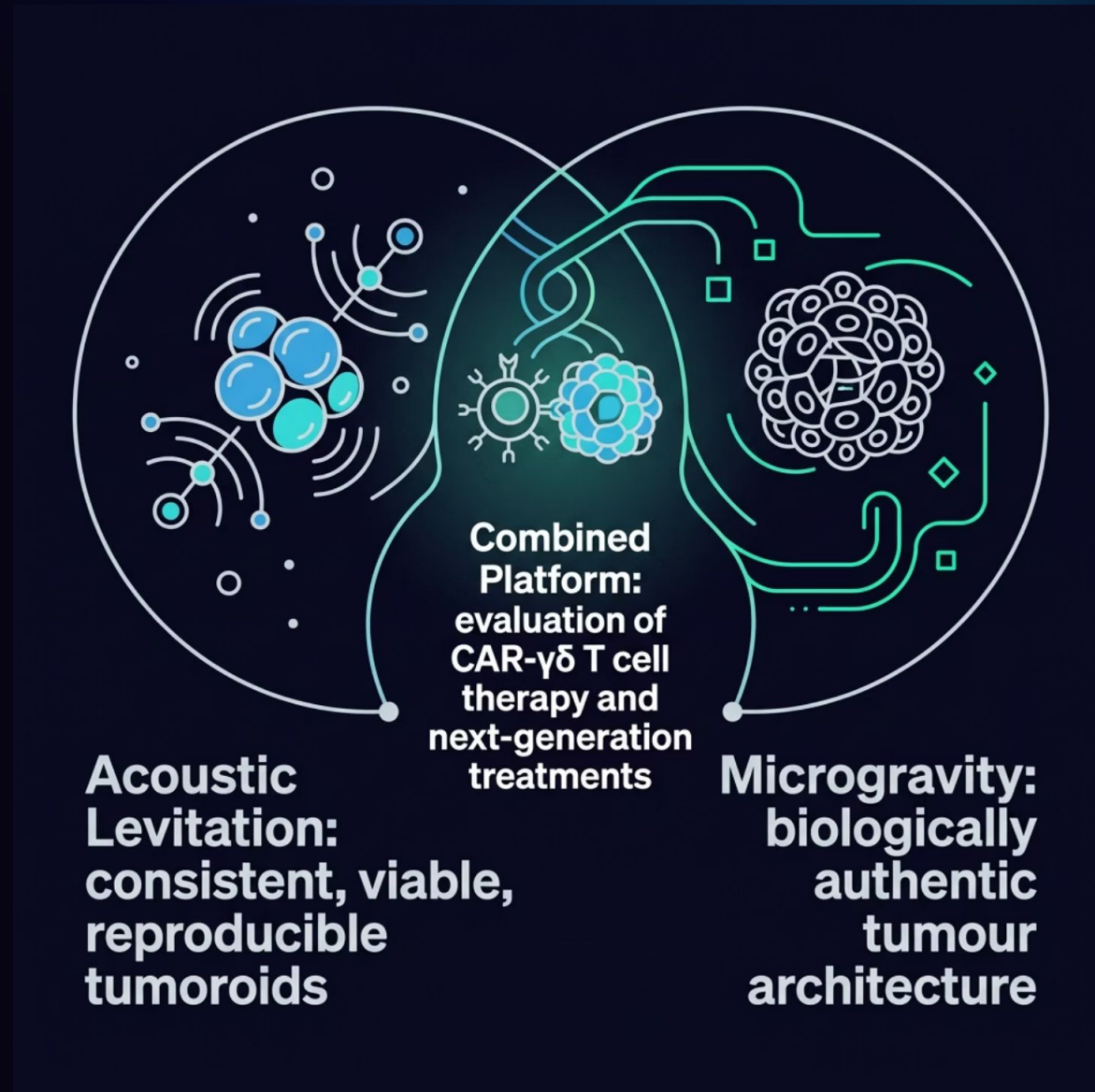
Hydrogel · Growth media · Drug delivery · 2× imaging ports (light in/out)



Proprietary ML Algorithm

Real-time feedback loop – continuously optimises acoustic parameters

Synergistic Impact



Acoustic Levitation

Consistent, viable,
reproducible tumoroids –
every single time



Microgravity

Biologically authentic
tumour architecture –
faithful to human disease



Enables evaluation of **CAR- $\gamma\delta$ T cell therapy** – published
in *Nature Communications Biology* – the fastest, most
faithful drug screening platform physically possible.



Cancer Treatment, there's a new sheriff in town.

\$6.3B

Market by 2030
Global cancer drugs market
— an industry ready for
disruption

10×

Faster Screening
More faithful models
accelerate the path from
discovery to clinic

1st

Of Its Kind
No comparable
acoustic-microgravity
disease modelling platform

Transform Research Deeper mechanistic understanding · Faster breakthroughs	Empower Therapies More effective · More targeted · More human
Save Lives For every patient like Martin who deserves more than 4 months	